YIFAN (JOHNNY) YANG

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EDUCATION

Indiana University, Bloomington, *August 2019 — August 2025*
Ph.D. in Mathematics
Advisor: Prof. Louis (Wai-Tong) Fan

University of Arizona *August 2015–May 2018*
B.S. in Mathematics, Magna Cum Laude
Minor in Finance

EMPLOYMENT AND ACADEMIC POSITIONS

Postdoctoral Research Scholars, Arizona State University, AZ *August 2025 — present*

RESEARCH INTERESTS

Probability theory and its applications, stochastic analysis, stochastic partial differential equations, Dirichlet forms, interacting particle systems, stochastic quantization, mathematical biology.

HONORS AND AWARDS

- William P Ziemer Award, Indiana University Bloomington, 2024.
- College of Arts and Sciences Fellowship, Indiana University Bloomington, 2020.
- Excellence in Undergraduate Research Award, Department of Mathematics, University of Arizona, 2018.

PUBLICATIONS AND WORK IN PROGRESS

Work in Progress

1. José Luis Pérez and Johnny (Yifan) Yang. Non-local dirichlet forms and stochastic quantization equations on metric measure spaces. In preparation, 2026+
2. Louis Fan, Johnny (Yifan) Yang, and John Yin. Spatial modeling for virus infection with multiple stages. In preparation, 2026+
3. Louis Fan, Adrián González Casanova, and Yifan (Johnny) Yang. Universality of stochastic fkpp equation. In preparation, 2026+

Preprints

1. Louis Wai-Tong Fan, Zhenyao Sun, and Johnny (Yifan) Yang. Comparison principles for stochastic reaction-diffusion equations on metric measure spaces. *arXiv preprint arXiv:2610.07790*, 2026. 100 pages
2. Adrián González Casanova and Johnny (Yifan) Yang. From individual-based models to general stochastic reaction diffusion equations. *arXiv preprint arXiv:2610.02669*, 2026. 54 pages
3. José Chacón, Adrián González Casanova, Imanol Nuñez, Rafael Peña Miller, José Luis Pérez, and Johnny (Yifan) Yang. Dormancy stabilizes non-transitive competitive dynamics. *arXiv preprint arXiv:2608.17179*, 2026. 20 pages
4. Hongyi Chen and Yifan (Johnny) Yang. Wick renormalized parabolic stochastic quantization equations on rough metric measure spaces. *arXiv preprint arXiv:2605.05442*, 2026. 71 pages

Published

1. Qiantong Liang, Johnny (Yifan) Yang, Wai-Tong Louis Fan, and Wing-Cheong Lo. Patch formation driven by stochastic effects of interaction between viruses and defective interfering particles. *PLOS Computational Biology*, 19(10), 2023
2. Wai-Tong Louis Fan, Yifan Johnny Yang, and Chaojie Yuan. Constrained langevin approximation for the togashi-kaneko model of autocatalytic reactions. *Mathematical Biosciences and Engineering*, 20(3):4322–4352, 2023

OTHER RESEARCH ACTIVITIES

Collaboration with John Yin (University of Wisconsin—Madison) under National Science Foundation (NSF) funding DMS-2348164, Fall 2024 — present.

- We aim to develop a spatial stochastic model for virus growth with mutations on two-dimensional domains.

Extinctions in multi-species stochastic systems, Indiana University, Fall 2023 — present
Collaboration with Louis Fan and John Yin

- We aim to study the extinction phenomenon in multi-species biological systems caused by stochastic effects.

Summer school Stochastic Quantization, Simons Laufer Mathematical Sciences Institute, Summer 2024.

PRESENTATIONS

Talks

- Population models and their SPDE limits, AMS Fall Western Sectional Meeting, Arizona State University, Tempe, AZ, November 8, 2026.
- Singular SPDEs on Rough Spaces, Aarhus Summer School in Analysis and Probability, Aarhus University, July 2026.
- Stochastic Quantization Equations on Metric Measure Spaces in the Da Prato–Debussche Regime, Probability and SPDEs Conference, Ohio State University, June 2026.
- Φ^4 on Rough Spaces: Constructing a Quantum Field Measure Beyond Euclidean Geometry, Postdoc seminar, Arizona State University, February 2026.
- Spatial stochastic modeling, CME seminar, Biodesign, Arizona State University, February 2026.
- Spatial stochastic models for population dynamics, Symposium of Probability and Stochastic Processes, Universidad de Sonora, November 2025.
- SPDEs on metric measure spaces, SPDEs Workshop at OSU, June 2025.
- SPDEs on spatial dimensions $d \in [1, 2 + \epsilon)$, Analysis Seminar, University of Illinois Chicago, March 2025.
- SPDEs on spatial dimensions $d \in [1, 2 + \epsilon)$, Probability Seminar, Arizona State University, February 2025.
- Spatial stochastic modeling for population dynamics, Population Genetics Seminar, University of Chicago, February 2025.
- Patch formation driven by stochastic effects, SIAM Conference on Mathematics of Data Science (MDS24), October 2024.
- Stability of Markov Process and Foster-Lyapunov Type Criteria Demonstration via Togashi-Kaneko Model, Graduate Probability Seminar, Indiana University Bloomington, September 2023.
- On Extinction Probability of Stochastic PDE on Metric Graphs, Graduate Student Probability Conference, University of Wisconsin–Madison, September 2022.

Poster Presentations

- Singular SPDE on metric measure spaces, Heat Kernels and Stochastic Analysis, Aarhus University, June–July 2026.
- Constructive QFT on rough spaces, Seminar on Stochastic Processes, Union College, March 2026.
- SPDEs on metric measure spaces, Seminar on Stochastic Processes, Indiana University Bloomington, March 2025.
- SPDE on metric measure spaces, Recent Progress in Stochastic Analysis and its Applications, Loyola University Chicago, July 2024.
- On Extinction Probability of Stochastic PDE on Metric Graphs, Seminar on Stochastic Processes, University of Arizona, April 2023.
- On Extinction Probability of Stochastic FKPP on Metric Graphs, Seminar on Stochastic Processes, Lehigh University, April 2022.

PROFESSIONAL ACTIVITIES

Organizer, Special Session on Mathematical Physics, Partial Differential Equations, and Probability, AMS Fall Western Sectional Meeting, Arizona State University, November 7–8, 2026.
Organizer, Graduate Probability Seminar at Indiana University Bloomington, Fall 2023 — Spring 2024.

TEACHING EXPERIENCE

Teaching

- STP 421: Probability, Fall 2026, Arizona State University.
- MAT 275: Modern Differential Equations, Spring 2026, Arizona State University.
- MAT 243: Discrete Mathematical Structures, Fall 2025, Arizona State University.

Mentoring, Indiana University, Bloomington

- Co-mentored an undergraduate woman for "Emerging Scholars Research Experience for Undergraduate Women," a year-long program at IU (August 2022–May 2023).
- Co-mentored an undergraduate for the Summer REU at Indiana University — Bloomington (Summer 2023).